

COUNTDOWN TIMER & STOPWATCH

WEB PROGRAMMING PROJECT

A MINI-PROJECT REPORT

Submitted by

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BACHELOR OF TECHNOLOGY

IN

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RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

An Autonomous Institution
CHENNAI

BONAFIDE CERTIFICATE

Certified that the project entitled "**COUNTDOWN TIMER & STOPWATCH**", submitted as a CS23531 – Web Programming Project, is the bonafide work of **KAVEEN KUMAR CS - (2116241001109)**, who carried out the work under my supervision.

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This mini-project report is submitted for the viva voce examination of CS23531 – Web Programming held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Accurate time measurement is a basic requirement in education, sports, cooking, laboratory work and everyday task management. This project presents a web-based **Countdown Timer and Stopwatch** application that gives users both tools through a single, lightweight and responsive interface. The countdown module lets the user set hours, minutes and seconds, choose quick presets, pause, resume and reset the timer, and be notified by a visual and audible alert when time expires. The stopwatch module measures elapsed time to the centisecond and records lap and split times, highlighting the best and worst laps.

The application is built with HTML for structure, CSS for styling and JavaScript for behaviour. Timing is based on system clock timestamps rather than simple interval counting, which keeps the displayed time accurate even when the browser throttles background tabs. A circular progress ring, a light and dark theme, and a mobile-friendly layout improve usability across devices. No installation, server or database is required, so the application can be hosted free of cost on platforms such as GitHub Pages.

By combining a clean user interface with reliable timing logic, the system aims to be simple, fast and accessible. The proposed system also provides a foundation for future improvements such as saved presets, multiple simultaneous timers, notifications, offline support through a progressive web app, and cloud-based storage of lap history.

ACKNOWLEDGEMENT

First, we thank the almighty God for the successful completion of the project. Our sincere thanks to our chairman **Mr. S. Meganathan**, B.E., F.I.E., for his sincere endeavor in educating us in his premier institution. We would like to express our deep gratitude to our beloved Chairperson **Dr. Thangam Meganathan**, for her enthusiastic motivation which inspired us a lot in completing this project and Vice-Chairman **Mr. Abhay Shankar Meganathan**, B.E., M.S., for providing us with the requisite infrastructure.

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KAVEEN KUMAR CS

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CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

The Countdown Timer and Stopwatch web application provides two essential time-tracking tools in one place. The countdown timer counts down from a user-defined duration and alerts the user at zero, while the stopwatch counts upward from zero and allows laps to be recorded. Both tools run entirely in the browser and require no installation.

1.2 SCOPE OF THE WORK

The system supports a configurable countdown timer with presets, pause and resume, a stopwatch with lap recording and best/worst lap indication, a light and dark theme, and a responsive layout for desktops, tablets and phones. It is intended for students, teachers, athletes, cooks and professionals who need quick and accurate timing without installing additional software.

1.3 PROBLEM STATEMENT

Many available timers are bundled with advertisements, require installation, or provide only one function. Physical stopwatches cannot store lap data, and basic online timers often drift when the browser tab is inactive. A single, free, ad-free and accurate web portal that provides both a countdown timer and a stopwatch can save time and avoid switching between several applications.

1.4 AIM AND OBJECTIVES OF THE PROJECT

The project aims to design and develop an accurate, responsive and user-friendly web application for time measurement. The objectives are:

- to implement a countdown timer with adjustable hours, minutes and seconds;
- to implement a stopwatch with centisecond precision and lap recording;
- to give clear visual and audible feedback when the countdown ends;
- to provide a responsive interface with light and dark themes; and
- to make the application deployable on free static hosting such as GitHub Pages.

CHAPTER 2
SYSTEM SPECIFICATIONS

2.1 HARDWARE SPECIFICATIONS

Processor	Intel i5
Memory	8 GB minimum
Storage	1 TB minimum

2.2 SOFTWARE SPECIFICATIONS

Operating system	Windows 10
Front-end	HTML5, CSS3, JavaScript (ES6)
Web browser	Google Chrome / Microsoft Edge / Firefox
Code editor	Visual Studio Code
Version control and hosting	Git and GitHub Pages
Data storage	Not required (runs in browser memory)

CHAPTER 3
MODULE DESCRIPTION

3.1 HOME AND NAVIGATION MODULE

The home module is the landing page of the application. It displays the welcome banner, a navigation bar and two cards that lead to the Countdown Timer and the Stopwatch. Navigation is handled in JavaScript by showing only the selected section, so switching between tools is instant and does not reload the page.

3.2 COUNTDOWN TIMER MODULE

This module accepts hours, minutes and seconds, validates them, and counts down to zero. It calculates the end time from the system clock and recomputes the remaining time on every tick, which prevents drift. A circular progress ring shows the fraction of time left. The user can start, pause, resume and reset the timer, or choose a preset of 1, 5, 10 or 25 minutes. When the timer reaches zero, the display blinks in red, a message is shown, the browser tab title changes, and a three-beep alert is generated using the Web Audio API.

3.3 STOPWATCH MODULE

This module measures elapsed time in minutes, seconds and centiseconds using the high-resolution *performance.now()* timer. The Start button toggles to Stop while running. The Lap button records the split time (time since the previous lap) and the total time, and the lap table lists the latest lap first. Once more than two laps exist, the fastest lap is shown in green and the slowest in red. Reset clears the display and all laps.

3.4 THEME AND RESPONSIVE UI MODULE

This module provides a light and dark theme using CSS variables, which are switched by a single class on the page body. Flexbox layouts and media queries adapt the interface to mobile screens so that all controls remain easy to use on small devices.

CHAPTER 4
SCREENSHOTS

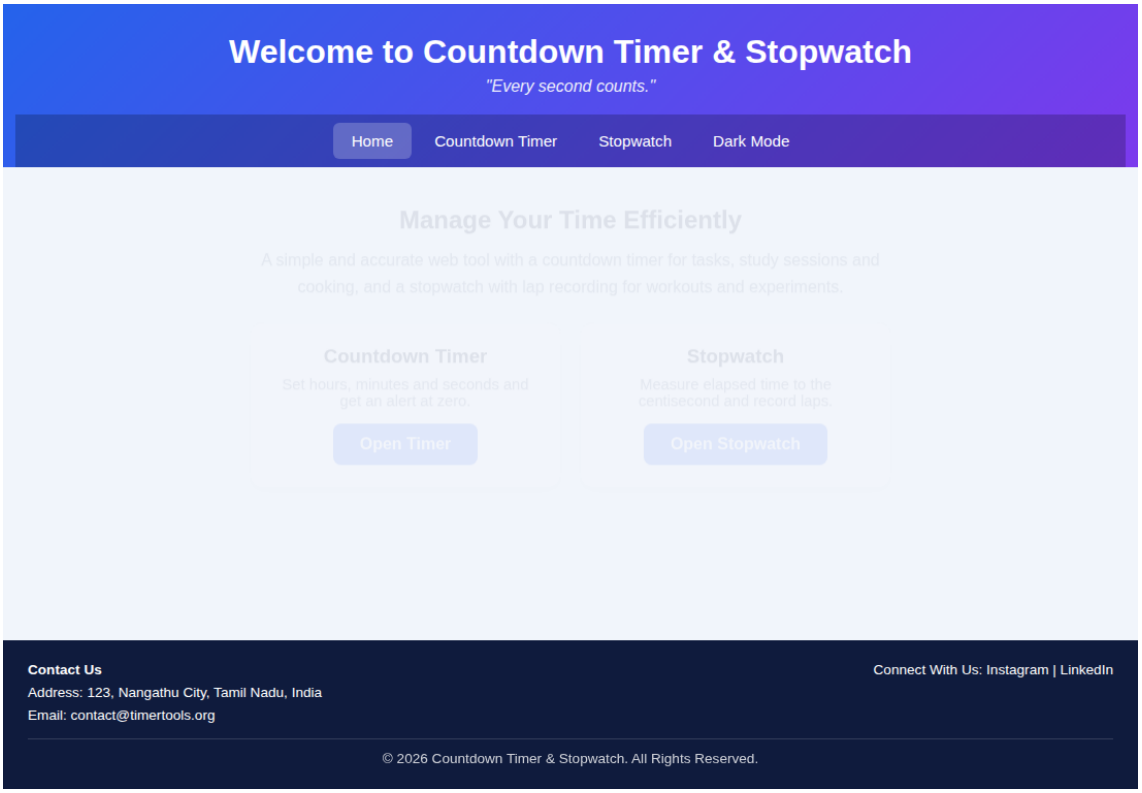


Figure 4.1: Introduction page

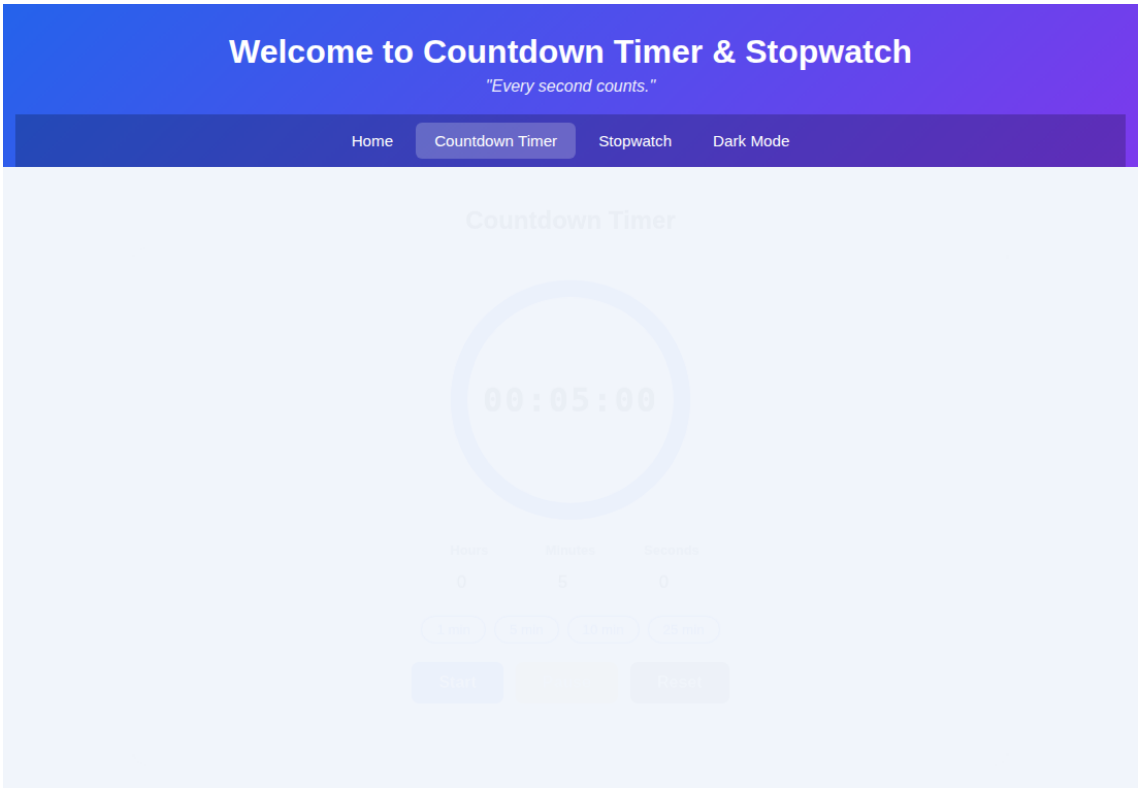


Figure 4.2: Countdown timer setup

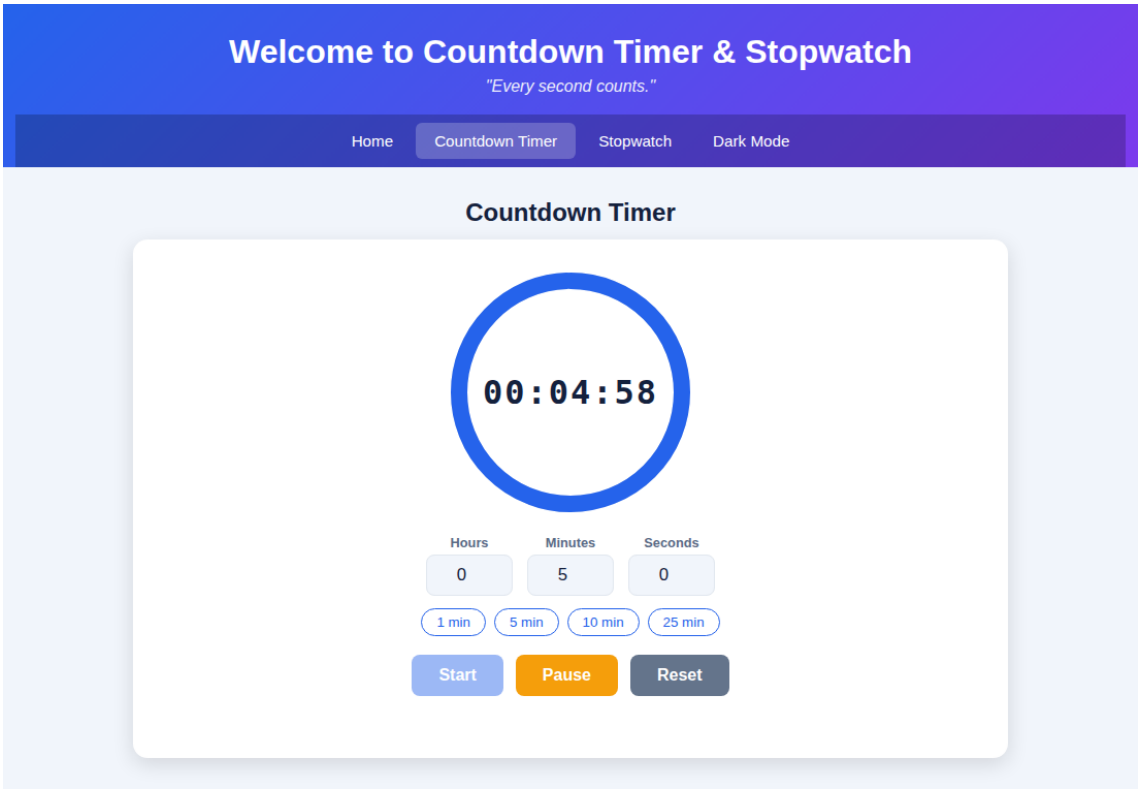


Figure 4.3: Countdown timer running

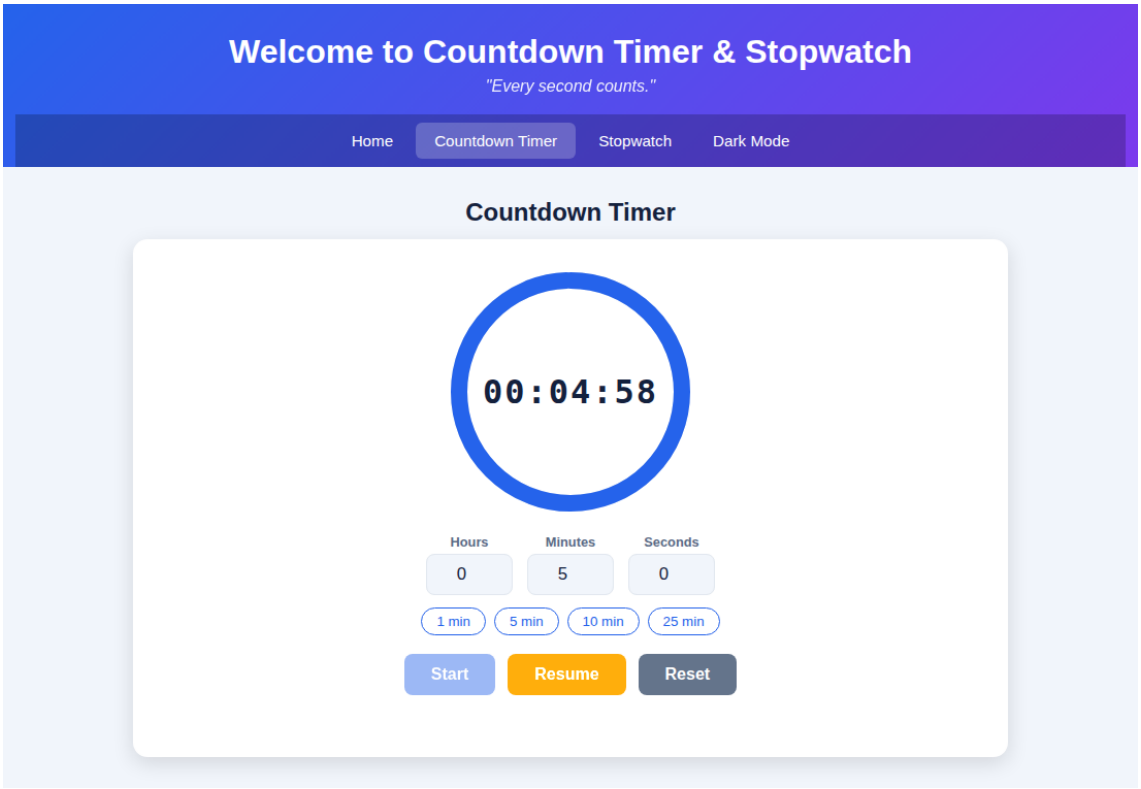


Figure 4.4: Countdown timer paused

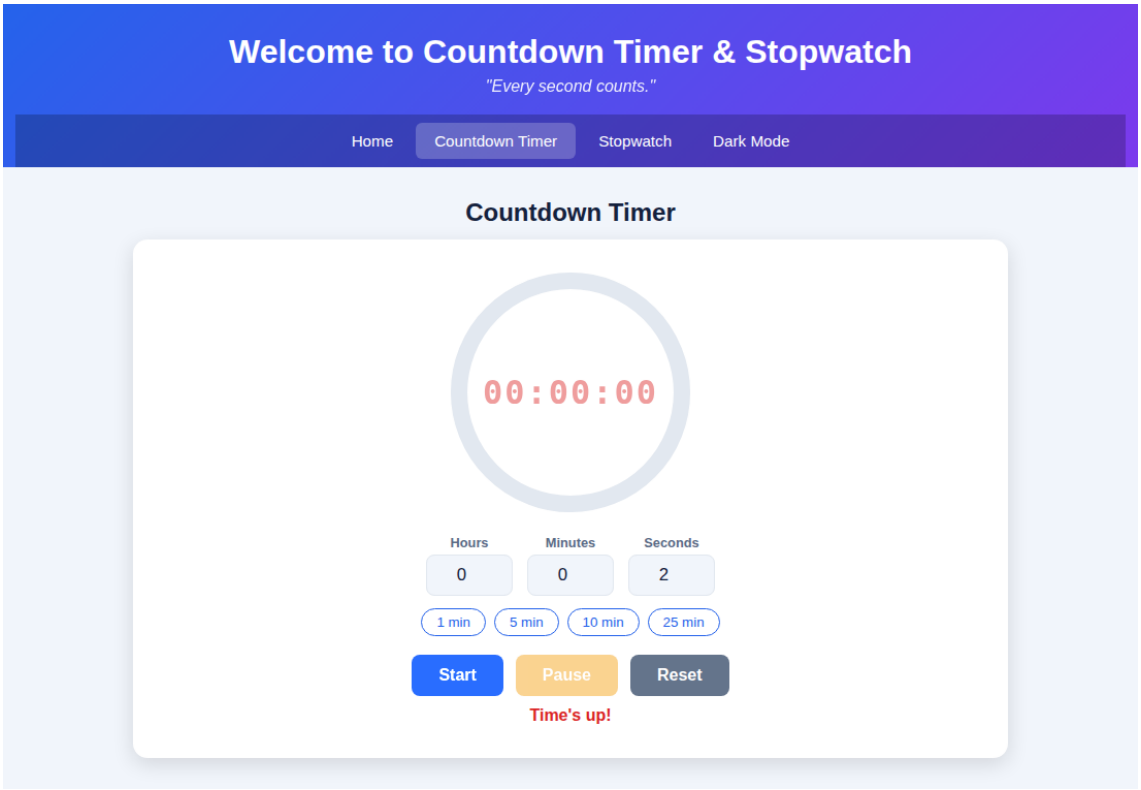


Figure 4.5: Countdown finished alert

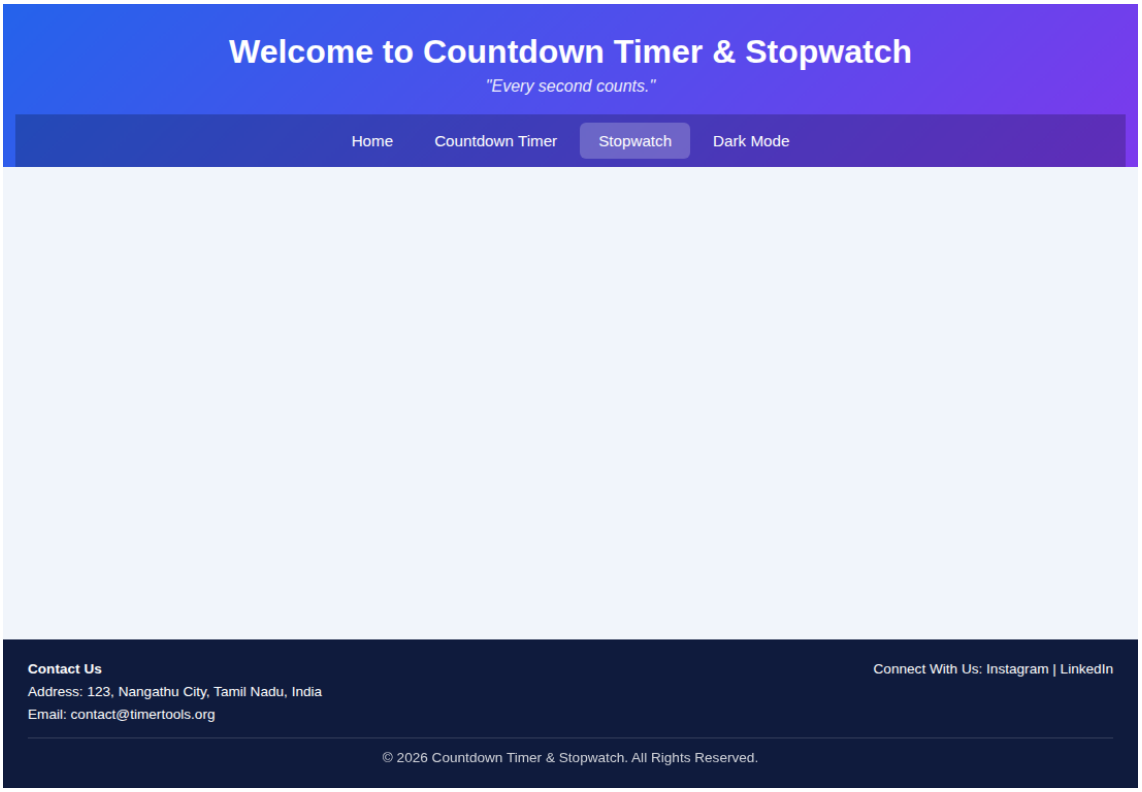


Figure 4.6: Stopwatch

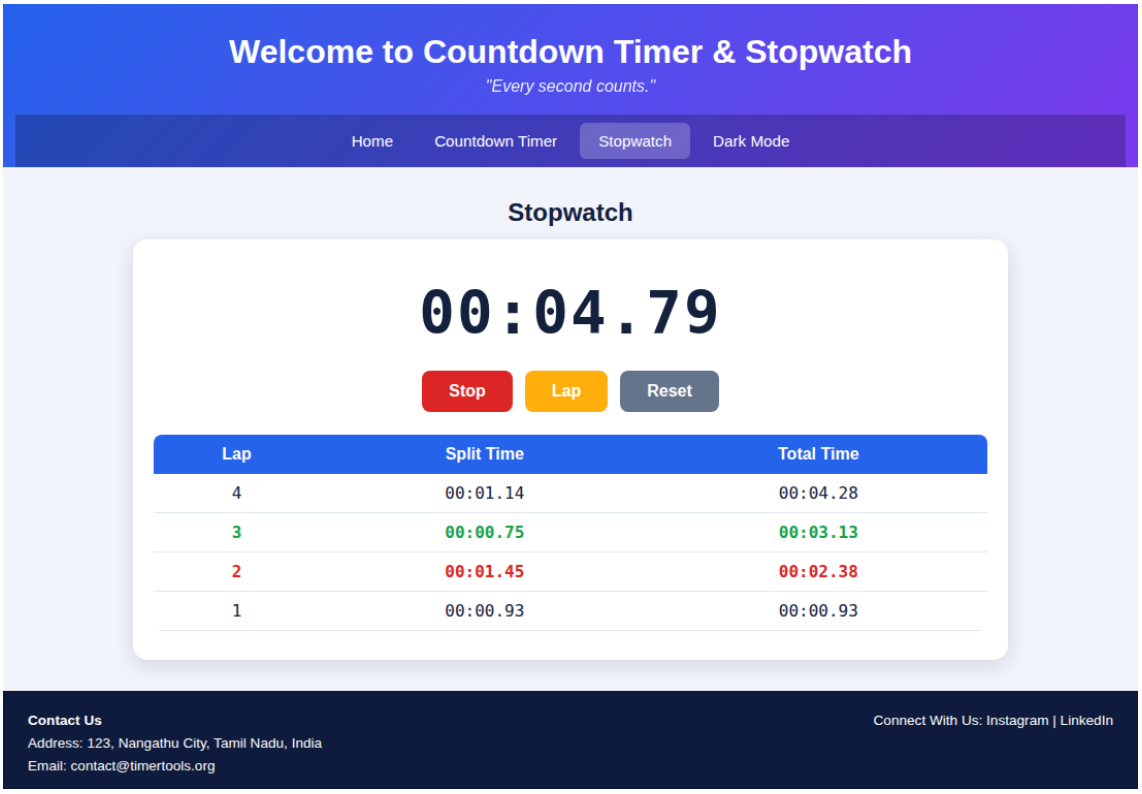


Figure 4.7: Stopwatch with lap records

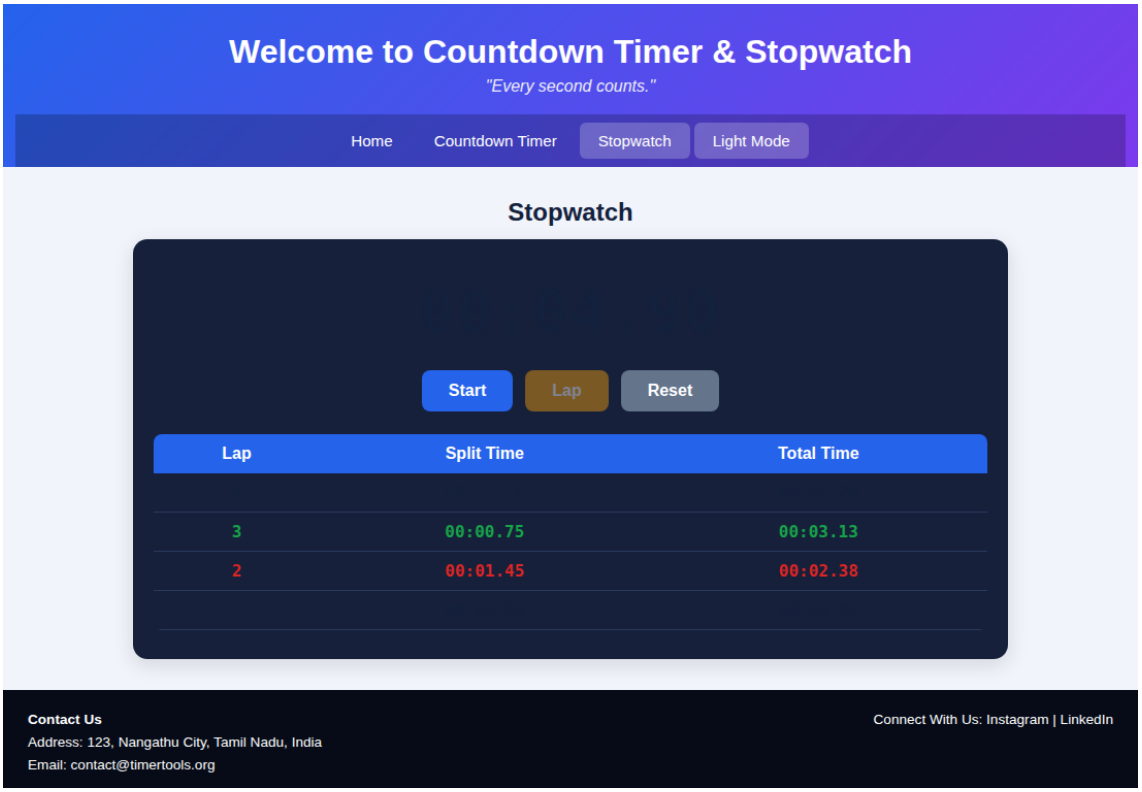


Figure 4.8: Dark theme

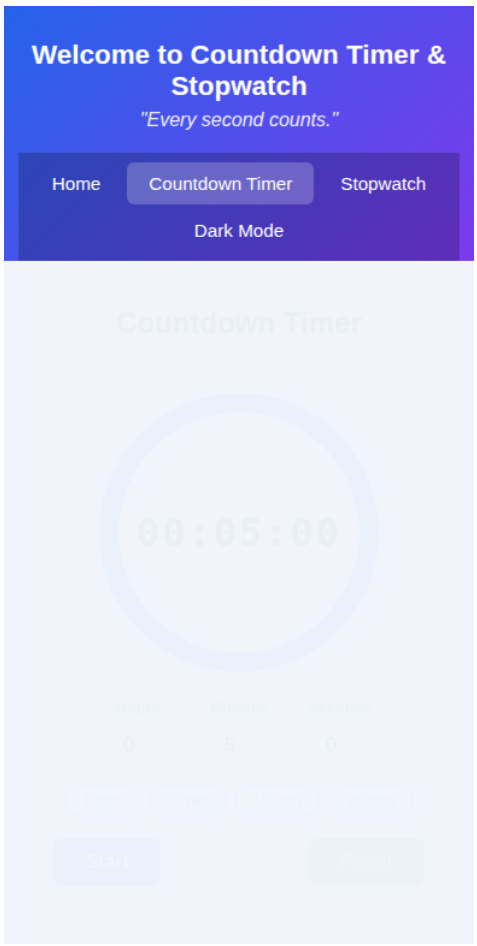


Figure 4.9: Mobile responsive view

CHAPTER 5
CONCLUSION AND FUTURE ENHANCEMENT

The Countdown Timer and Stopwatch web application provides an accurate, simple and accessible way to measure time. By using timestamp-based calculations it avoids drift, and its lap table, progress ring, alerts and themes make it practical for everyday use. Since it runs entirely in the browser, it is fast, free to host and works on all modern devices.

Future enhancements may include saving custom presets and lap history, running multiple timers at the same time, desktop and push notifications, exporting laps to CSV, offline support as a progressive web app, voice commands, and cloud-based storage with user accounts.

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